

Aryan Miriyala

+1 419-315-0444 | aryanmiriyala@gmail.com | [linkedin.com/in/aryan-miriyala](https://www.linkedin.com/in/aryan-miriyala) | github.com/aryanmiriyala

PROFESSIONAL SUMMARY

Software engineer with experience building reliable backend/data workflows, secure internal tools, and AI-assisted engineering systems using **Python, SQL, TypeScript, PostgreSQL, Oracle, AWS, and Docker**. Background includes production data automation, database validation, access control, reproducible developer environments, and communicating technical tradeoffs across operational systems.

EDUCATION

Bowling Green State University <i>Master of Science in Computer Science, GPA: 4.00/4.00</i>	Bowling Green, OH <i>Aug. 2024 – Aug. 2026</i>
Bowling Green State University <i>Bachelor of Science in Computer Science, GPA: 4.00/4.00</i>	Bowling Green, OH <i>Jan. 2021 – Apr. 2024</i>

EXPERIENCE

Artificial Intelligence & Computer Science Intern <i>SmartSolve Industries</i>	May 2026 – Present <i>Bowling Green, OH</i>
- Built secure internal software with Next.js, TypeScript, PostgreSQL, Drizzle ORM, SSO, and auth middleware for onboarding and org-chart workflows around sensitive employee data. - Created a cross-platform Docker devcontainer for WSL2/Colima, standardizing reproducible environments for AI-assisted development, debugging, implementation planning, and code review. - Applied Codex and Claude Code to accelerate system design and feature iteration while validating generated output through human review and proprietary-code isolation.	
Data Engineering Intern <i>American Association of Insurance Services</i>	May 2025 – Aug. 2025 <i>Lisle, IL</i>
- Automated a manual SQL billing workflow with Python, PySpark, and AWS Glue, processing 20+ TB of production insurance data for 700+ member companies. - Profiled 160+ Oracle/Impala tables with Python, Pandas, SQL, and JDBC to map source-system similarities and define 25 MDM domains for cloud ETL migration. - Built ETL validation, reverse-ETL, and custom AWS IAM access patterns across S3 and MDM workflows, improving controlled data access and reducing recurring manual operations.	
Software Engineering Intern <i>American Association of Insurance Services</i>	May 2023 – May 2024 <i>Lisle, IL</i>
- Engineered Python/JSON backend automation generating 1,000+ production SQL tables across Oracle/PostgreSQL, supporting standardized data models across 10+ insurance lines. - Built Python/Pandas workflows with multiprocessing, fuzzy matching, and hashing to clean large CSV datasets for production data-standardization pipelines. - Contributed to a React/Node.js modernization by implementing RBAC and JWT patterns, replacing broad employee-wide access with role-specific controls for roughly 110 internal users. - Automated event-driven database validation with AWS Lambda triggers that tokenized PII using hashlib/boto3 and ingested structured JSON into S3.	
Software Engineering Intern <i>Alliance for Paired Kidney Donation</i>	May 2022 – Aug. 2022 <i>Toledo, OH</i>
- Secured an AWS-hosted Lucee/CFML healthcare workflow platform with global anti-CSRF protection, IP-aware audit logging, and access-denied/404 routing across form-heavy internal workflows. - Modernized HTML5 date/time inputs, logistics defaults, Matchgrid import ordering, MFI upload paths, and duplicate-antigen crash handling to improve data-entry reliability for internal operations.	

PROJECTS

HealthTrend Data Pipeline <i>Kafka, Hadoop HDFS, Spark, PySpark, Docker</i> : Built a containerized healthcare data pipeline routing CSV patient records and JSON health events into batch and streaming processing paths.
Self-Adaptive Parallelism <i>C++, OpenMP, Python, Docker, Linux</i> : Built a runtime research system using per-epoch telemetry and a UCB controller to tune scheduling policy, chunk size, and thread count.
RocketGrader <i>Angular, TypeScript, Express, MongoDB, Auth0, AWS S3</i> : Built a full-stack AI feedback platform with REST APIs, role-based dashboards, file upload/storage, parsing, and LangChain/Mistral AI grading workflows.
Diff-Grounded PR Generation <i>Python, OpenAI API, GitHub API</i> : Built an LLM pipeline generating evidence-grounded PR descriptions from commits, diffs, linked issues, and metadata with automated evaluation against code-change evidence.

TECHNICAL SKILLS

Languages: Python, SQL, TypeScript/JavaScript, C++, C#, C, Lucee/CFML
Backend, Databases & APIs: Node.js, Express, REST APIs, PostgreSQL, MySQL, Oracle, MongoDB, Drizzle ORM, JDBC
Cloud, Data & Systems: AWS Glue, Lambda, S3, IAM, PySpark, Pandas, Kafka, Hadoop HDFS, Docker, Git, Linux
Reliability, Security & AI: SSO, JWT, RBAC, Auth0, anti-CSRF, audit logging, PII tokenization, Codex, Claude Code, OpenAI API, Mistral AI, LangChain